
ML Factor Investing

Kickoff Plan — Day 1 & Day 2

Systematic Equity Factor Investing with Regime-Detection Overlay
Team of 3 • 5-Week Project

Project goal. Build an ML-based cross-sectional equity strategy on the S&P 500 (long-short portfolio), with a Gaussian Mixture regime-detection overlay that modulates leverage. Evaluate with a rigorous walk-forward backtest including transaction costs. Methodology inspired by Gu, Kelly & Xiu (2020).

This document. Everything the team needs to execute Day 1 and Day 2. Assumes scope, roles, and communication channels are already agreed.

1. GitHub Repo Setup

Do this together, screen-share, ~30 minutes.

1. Go to github.com. Person A creates a new repository named `ml-factor-investing-2026`. Keep it **private** for now; flip to public once polished.
2. Initialize with a README, a `.gitignore` (pick *Python* from GitHub's dropdown), and a license (MIT is fine).
3. Person A adds Persons B and C as collaborators: *Settings* → *Collaborators* → *Add people*.
4. All three clone locally:

```
git clone https://github.com/[username]/ml-factor-investing-2026.git
cd ml-factor-investing-2026
```

5. Create the folder structure. Person A runs these commands and commits:

```
mkdir data src notebooks results report
touch src/__init__.py
touch src/data_loader.py src/factors.py src/models.py
touch src/backtest.py src/regime.py src/metrics.py
touch notebooks/01_data_exploration.ipynb
touch requirements.txt
```

6. Populate `requirements.txt`:

```
pandas
numpy
scikit-learn
xgboost
torch
hmmlearn
yfinance
pandas-datareader
```

```
matplotlib
seaborn
shap
optuna
jupyter
```

7. Agree on the git workflow:

- Never push directly to `main`.
- Each person works on their own branch (`persona-data-pipeline`, `personb-models`, `personc-regime`).
- Open a pull request to merge into `main`; at least one other person reviews before merging.
- Commit messages in present tense (“*add factor construction*”, not “*added*”).

2. Google Colab Familiarization

Individually, ~45 minutes each. Laptops are your main environment; Colab is the backup for the biggest training run.

1. Go to `colab.research.google.com`, sign in.
2. New notebook. Run `print("hello")` to confirm it works.
3. *Runtime* → *Change runtime type* → *Hardware accelerator* = *T4 GPU* (free tier).
4. Confirm GPU:

```
import torch
print(torch.cuda.is_available()) # should print True
```

5. Mount Google Drive for persistent storage:

```
from google.colab import drive
drive.mount('/content/drive')
```

6. Try a full workflow once: `git clone` the repo inside Colab and run a script. One person does this so all kinks are known.

Free Colab limits: ~12 hours continuous GPU/day, disconnects after ~90 min idle, GPU availability not guaranteed. Adequate for your needs — just never keep the *only copy* of anything on Colab.

3. Understanding the Core Goal — Reading Plan

Everyone reads item 1. The rest is role-specific.

Everyone (today, ~90 minutes)

Read the abstract and introduction of **Gu, Kelly & Xiu (2020)** — *Empirical Asset Pricing via Machine Learning, Review of Financial Studies*. Free PDF via Google. **Pages 1–10 only.** You’re after three things:

- What problem are they solving? (Predicting stock returns with ML.)

- What's the setup? (Monthly prediction of next-month returns, many stocks, many features.)
- What do they find? (Tree-based methods and neural nets beat linear models; momentum and liquidity features dominate.)

This paper is the north star of the project. Every design decision will reference it.

Person A — Data & Infrastructure

Focus: data pitfalls and backtest methodology. Become the team expert here.

Reads (today/tomorrow, ~3h):

1. *Advances in Financial Machine Learning* by Marcos López de Prado, Chapter 11 (“Back-testing”). University library likely has the book. Understand walk-forward vs. standard k -fold CV, and why normal CV is disastrous for time-series finance.
2. A blog post on “*Survivorship Bias in Financial Data*” (~15 min). The single most important data pitfall.

Lectures to re-watch:

- Lecture 4 (Overfitting, Bias-Variance, Regularization) — cross-validation especially.
- Lecture 2 (Regression) — the OLS foundation.

Today's specific task: Skim the yfinance documentation and `pandas-datareader` / FRED docs. **By end of Day 2:** pull S&P 500 price history for 10 stocks into a DataFrame.

Person B — Alpha Model

Focus: factor construction and ML model selection for finance. Become the team expert here.

Reads (today/tomorrow, ~3h):

1. **Gu, Kelly & Xiu (2020)**, **Sections 2 & 3** in detail. Section 2: the 94 features they use. Section 3: the models. You won't use all 94 features; you need to understand how they're constructed.
2. **Fama & French (1993)** — the original three-factor paper. Skim only. You're learning the intellectual lineage: size, value, momentum were the original “factors”; you're extending with ML.
3. **XGBoost tutorial:** xgboost.readthedocs.io/en/stable/tutorials/model.html (~45 min, read cover to cover).

Lectures to re-watch:

- Lecture 2 (Regression) — non-linear basis functions.
- Lecture 4 (Regularization) — Lasso is your linear baseline.
- Lecture 8 (Decision Trees & Ensembles) — the foundation for XGBoost.
- Lectures 5–6 (Neural Networks) — for the NN baseline.

Today's specific task: Read the Gu-Kelly-Xiu feature table. Pick the 5 factors we will use. Write them in the shared doc with *exact* formulas.

Person C — Risk & Regime

Focus: regime detection. Become the team expert here.

Reads (today/tomorrow, ~2h):

1. **Nystrup, Madsen & Lindström (2018)** — *Dynamic Allocation or Diversification: A*

Regime-Based Approach to Multiple Assets. Search the title; available online. Skim — focus on how they define and use regimes.

2. A tutorial on GMMs applied to finance. Search: “*Gaussian Mixture Models market regimes python*”. Pick a Medium/blog post with working code.

Lectures to re-watch:

- Lecture 10 (Clustering: k -Means, GMMs) — core toolkit.
- Lecture 11 (Generative Models — GMM + EM) — deepens understanding.
- Lecture 9 (PCA) — for dim-reduction on macro features if needed.

Today’s specific task: By end of Day 2, write a one-pager in the shared drive titled “*Regime Detection: Approach and Features*” listing the 5 features you’ll use and why. This is the foundation of your workstream.

4. What to Actually *Do* Today

Aim to finish before bed.

Everyone

- Create GitHub, Google Colab, and FRED accounts.
- Clone the repo. Run `pip install -r requirements.txt` in a virtual environment. Confirm everything installs.
- Start the role-specific reading.

Person A

- Write the first draft of the backtest function signature in `src/backtest.py` (docstring + function skeleton only, no implementation). Push to a branch, open a PR for discussion.
- Pull one stock’s price history with `yfinance` to confirm the environment works.

Person B

- Open a notebook, import `xgboost`, train it on `sklearn` iris just to confirm end-to-end plumbing.
- Start reading Gu, Kelly & Xiu properly.

Person C

- Open a notebook, import `sklearn.mixture.GaussianMixture`, fit it to a toy 2D dataset from `sklearn.datasets.make_blobs`, plot the result. Confirms the setup works.
- Start reading on regime detection.

5. What to Do Tomorrow (Day 2)

- **Morning.** Finish your reading. Come to the 15-min standup ready to summarize what you learned in 2 minutes each.
- **Afternoon.** Person A starts pulling real data. Person B finalizes the factor list. Person C finalizes the regime feature list.
- **End of day.** Push what you have to your branches. Merge setup code into `main`.

By end of Day 2 you should have: working dev environments for all 3, a populated GitHub repo,

aligned understanding of the project, and a clear factor + regime feature list.

6. Principles (Agree Today — Saves Days Later)

1. **One language, one style.** Python 3.11, PEP 8, auto-format with `black`.
2. **Notebooks are for exploration only.** All reusable code lives in `src/*.py`. No production code in notebooks. Prevents the classic *“it worked in Person B’s notebook but Person A can’t reproduce it”* disaster.
3. **Lock the data.** Once pulled, save as parquet files in `data/` (gitignore the folder itself, but commit the download script). Nobody re-hits yfinance/FRED repeatedly.
4. **Random seeds everywhere.** Every model gets `random_state=42`. Every sampling step gets a seed. Reproducibility is non-negotiable — you need it for the report.
5. **Keep a decisions log.** A `DECISIONS.md` file in the repo where you log every significant choice (*“S&P 500 over Russell 1000 because...”*). The report basically writes itself from this in Week 5.

7. Minimum Deliverable — End of Day 2

If someone asked *“what did you do in the first two days?”* you should be able to show:

- A working, populated GitHub repo with the agreed folder structure.
- Every team member’s dev environment running: `import xgboost; import hmmlearn; import yfinance` all succeed.
- A 1-page shared doc summarizing project goal, factors, regime features, and roles.
- One stock’s historical data pulled successfully into a `DataFrame`.
- Three completed reading assignments, with key takeaways in the shared doc.

Hit this checklist and you are perfectly on track for the 5-week plan.

Good kickoff = half the battle. Ship it.